# Casimir torque between two birefringent plates

Cheng-Gang Shao, Ai-Hong Tong, and Jun Luo*
*Department of Physics, Huazhong University of Science and Technology, Wuhan 430074, People's Republic of China*
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The vacuum energy between two birefringent plates was calculated using the quantized surface mode technique. And an analytical approximate expression of Casimir torque was obtained in the retarded limit, which is found to vary as $\sin(2\varphi)$ with $\varphi$, the angle between the two optical axes. This result is a supplement to the numerical calculation obtained by Munday, Iannuzzi, Barash, and Capasso [Phys. Rev. A. **71**, 042102 (2005)].

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## I. INTRODUCTION

During the last few years the Casimir effect has attracted a lot of experimental and theoretical attention as a nontrivial macroscopic evidence for the existence of zero-point oscillations of electromagnetic field [1–8]. This is in part due to the resurgence of the interest in the fundamental problems connected with the physical vacuum. The electromagnetic vacuum pressure is the most popular interpretation of the Casimir effect [9]. Since photons also carry the angular momentum, the vacuum torque will appear between macroscopic bodies when their characteristic properties are anisotropic [10–13].

Lifshitz first introduced a macroscopic theory for the calculation of the Casimir force between two dielectric plates [14], which was developed by many authors, such as Schwinger [15,16], Matloob [17–19], and Tomas [20]. One of the limitations of the Lifshitz theory is the assumption that the dielectric properties of the interacting materials are isotropic. For the anisotropic material, Parsegian and Barash have discussed the Casimir effect between two dielectrically anisotropic plates immersed in a third anisotropic material [21,22]. As a theoretical derivation, the torque between two birefringent plates in a one-dimension calculation has been presented [10]. Recently, the numerical calculation of the Casimir torque on birefringent plates was presented, and a possible experiment was also proposed [23].

In this paper, based on the quantized surface mode technique [1,24,25], we rederive the Casimir torque on two parallel birefringent plates with in-plane optical anisotropy, separated by vacuum. An analytical approximate expression of Casimir torque was obtained in the retarded limit, which is found to vary as $\sin(2\varphi)$ with $\varphi$, the angle between the two optical axes. This result is consistent with that obtained by Refs. [10,23].

## II. QUANTIZED SURFACE MODES

We will consider two plates (diameter $D$, thickness $d$) made of the same uniaxial medium kept parallel at a distance $a$. The two optical axes are in the $x$-$y$ plane and the $z$ axis is orthogonal to the plates. $\varphi$ is the angle between two optical axes (as shown in Fig. 1). In view of $D \gg a$, $d \gg a$, we may consider two semi-infinite plates with in-plane optical axis oriented as shown in Fig. 2. The three regions have the permittivity tensors $\vec{\varepsilon}_1$, $\varepsilon_3$, and $\vec{\varepsilon}_2$ respectively. The region between the two plates is assumed vacuum.

The vacuum fluctuation fields can be determined for the system defined above. For our purposes we need consider only the zero-point energy associated with the surface modes which are exponentially decaying for $z<0$ and $z>a$. In general, the electric and the magnetic fields of the surface mode $q$ can be written in the forms [24]

$$\vec{E}_q = iNe^{ik_x x+ik_y y}(a_q \vec{e}_q e^{ik_z z}e^{-i\omega t} - a_q^\dagger \vec{e}_q^{\,*} e^{-ik_z z}e^{i\omega t}), \qquad (1)$$

$$\vec{H}_q = Ne^{ik_x x+ik_y y}(a_q \vec{h}_q e^{ik_z z}e^{-i\omega t} + a_q^\dagger \vec{h}_q^{\,*} e^{-ik_z z}e^{i\omega t}), \qquad (2)$$

because of the uniform in each region. $N=(2\pi\hbar\omega/L^2)^{1/2}$ is the normalization factor, with $L$ the length of the normalization box in the $x$ and $y$ direction, and $\vec{k}_\parallel=(k_x,k_y)$ is the propagation vector in the $x$-$y$ plane. $a_q^\dagger$ and $a_q$ are the usual creation and annihilation operators, respectively. In order to determine mode of the electric $\vec{e}_q$ and the magnetic fields $\vec{h}_q$ with frequency $\omega$, we need to consider the classical Maxwell

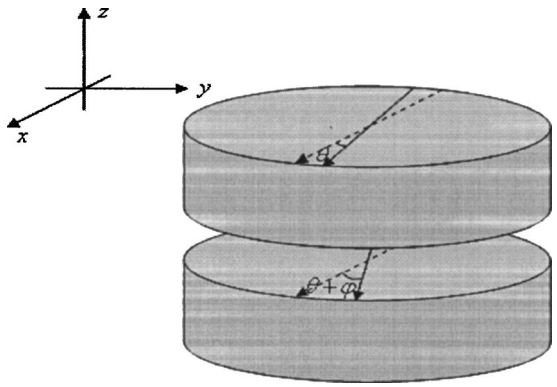



FIG. 1. Sketch of the system under investigation. Two uniaxial parallel plates with in-plane optical axis are placed in close proximity. The $z$ axis is chosen to be orthogonal to the plates. $\theta$ and $\theta+\varphi$ are the angles between $x$ axis and optical axes, with $\varphi$ the angle between two optical axes.

*Electronic address: junluo@mail.hust.edu.cn

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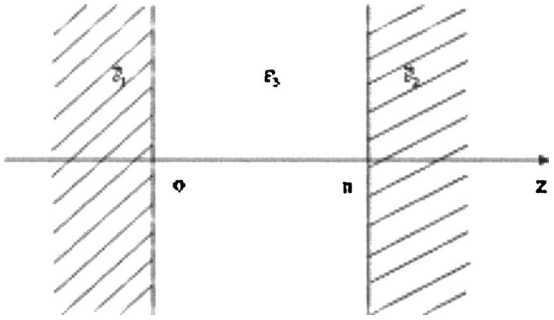



FIG. 2. Configuration of two parallel semi-infinite anisotropic dielectric plates. The $z$ axis is orthogonal to the plates and two optical axes are in the $x$-$y$ plane.

equations without sources in the uniaxial medium,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t}\vec{B},$$

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2}\frac{\partial}{\partial t}\overleftrightarrow{\varepsilon}\vec{E}. \tag{3}$$

For simplicity of the mode description, we choose the coordinates system in which $\vec{k}_{\parallel}$ is parallel to the $x$ axis. So the wave vector $\vec{k}$ can be expressed as

$$\vec{k} = (k_x, k_y, k_z) = K_0(\alpha, 0, \gamma) \tag{4}$$

with $K_0 = \omega/c$, $K_{\parallel} = K_0\alpha$, and $K_{\perp} = K_0\gamma$.

The characteristic feature of a uniaxial medium is the existence of a distinguished axis (optical axis). If one of the coordinate axes is chosen to be parallel to this distinguished direction, the permittivity tensor will be diagonal and the element associated with the distinguished axis is different from the remaining two diagonal ones,

$$\overleftrightarrow{\varepsilon} = \varepsilon_{\perp}\overleftrightarrow{\mathbf{I}} + (\varepsilon_{\parallel} - \varepsilon_{\perp})\vec{c}\vec{c}, \tag{5}$$

where $\overleftrightarrow{\mathbf{I}}$ is the unit tensor and $\vec{c}$ is the unit vector in the direction of the optical axis. The subscripts $\parallel$ and $\perp$ indicate the values of the dielectric tensor along the optical axis and along the plane orthogonal to the optical axis. In our calculation, the two optical axes of two uniaxial media are in the $x$-$y$ plane and rotated by angles $\theta$ and $\theta+\varphi$ with respect to the $x$ axis. The dielectric tensors of two plates are then described by the following matrices:

$$\overleftrightarrow{\varepsilon}_1 = \begin{pmatrix} \varepsilon_{\parallel}\cos^2\theta + \varepsilon_{\perp}\sin^2\theta & (\varepsilon_{\parallel} - \varepsilon_{\perp})\sin\theta\cos\theta & 0 \\ (\varepsilon_{\parallel} - \varepsilon_{\perp})\sin\theta\cos\theta & \varepsilon_{\parallel}\sin^2\theta + \varepsilon_{\perp}\cos^2\theta & 0 \\ 0 & 0 & \varepsilon_{\perp} \end{pmatrix}, \tag{6}$$

$$\overleftrightarrow{\varepsilon}_2 = \begin{pmatrix} \varepsilon_{\parallel}\cos^2(\theta+\varphi) + \varepsilon_{\perp}\sin^2(\theta+\varphi) & (\varepsilon_{\parallel} - \varepsilon_{\perp})\sin(\theta+\varphi)\cos(\theta+\varphi) & 0 \\ (\varepsilon_{\parallel} - \varepsilon_{\perp})\sin(\theta+\varphi)\cos(\theta+\varphi) & \varepsilon_{\parallel}\sin^2(\theta+\varphi) + \varepsilon_{\perp}\cos^2(\theta+\varphi) & 0 \\ 0 & 0 & \varepsilon_{\perp} \end{pmatrix}. \tag{7}$$

According to Eqs. (4) and (6), and substituting $(\vec{E}, \vec{B}) = (e_x, e_y, e_z, -h_x, h_y, h_z)e^{iK_0(\alpha x + \gamma z) - i\omega t}$ into Eq. (3), we obtain the eigenequations of the transverse elements of the electromagnetic field mode $e_x$, $e_y$, $h_y$, $-h_x$ in the region I as the following:

$$\begin{pmatrix} 0 & 0 & -\frac{t^2}{\varepsilon} & 0 \\ 0 & 0 & 0 & 1 \\ \varepsilon(1+\delta\cos^2\theta) & \varepsilon\delta\sin\theta\cos\theta & 0 & 0 \\ \varepsilon\delta\sin\theta\cos\theta & \varepsilon\delta\sin^2\theta - t^2 & 0 & 0 \end{pmatrix} \begin{pmatrix} e_x \\ e_y \\ ch_y \\ -ch_x \end{pmatrix} = \gamma \begin{pmatrix} e_x \\ e_y \\ ch_y \\ -ch_x \end{pmatrix}, \tag{8}$$

where $t^2 = \alpha^2 - \varepsilon$ and $\varepsilon_{\perp} = \varepsilon$, $\varepsilon_{\parallel} = (1+\delta)\varepsilon_{\perp}$. We use the $4\times4$ matrix formulation as in Ref. [26] to calculate the wave vectors and the mode fields. And the four eigenvectors can be written in the matrix form

$$W=\begin{pmatrix} -\frac{i\sin\theta}{t} & -\frac{it^2\cos\theta}{\varepsilon t_1} & \frac{i\sin\theta}{t} & \frac{it^2\cos\theta}{\varepsilon t_1} \\ \frac{i\cos\theta}{t} & \frac{i\sin\theta}{t_1} & -\frac{i\cos\theta}{t} & -\frac{i\sin\theta}{t_1} \\ \frac{\varepsilon\sin\theta}{t^2} & \cos\theta & \frac{\varepsilon\sin\theta}{t^2} & \cos\theta \\ \cos\theta & \sin\theta & \cos\theta & \sin\theta \end{pmatrix} \quad (9)$$

with $t_1=\sqrt{t^2+t^2\delta\cos^2\theta-\varepsilon\delta\sin^2\theta}$. Every column of the $W$ matrix represents a eigenvectors. The corresponding eigenvalues are

$$\gamma\equiv[\gamma_1 \quad \gamma_2 \quad \gamma_3 \quad \gamma_4]=[-it \quad -it_1 \quad it \quad it_1]. \quad (10)$$

Four eigenvalues correspond to four independent mode solutions. The general solution of the electromagnetic field should be the linear superposition of the four mode solutions, and the superposition coefficients are the amplitudes of each mode. Then the transverse elements of the electromagnetic field mode in Eqs. (1) and (2) may be written as

$$\begin{pmatrix} e_x \\ e_y \\ ch_y \\ -ch_x \end{pmatrix}=W\begin{pmatrix} e^{iK_0(\alpha x+\gamma_1 z)}A_1 \\ e^{iK_0(\alpha x+\gamma_2 z)}A_2 \\ e^{iK_0(\alpha x+\gamma_3 z)}A_3 \\ e^{iK_0(\alpha x+\gamma_4 z)}A_4 \end{pmatrix}e^{i\omega t} \quad (11)$$

in region I, where $A_j$ is the amplitude of the $j$th mode solution. For region II, the transverse elements of the electromagnetic field mode have the same form as Eq. (11), except for $\theta$, $A_j$ being replaced by $\theta+\varphi$, $B_j$.

The vacuum region III is different from the regions I and II. The eigenvalues and eigenvectors can be obtained in the same way as

$$\gamma^0=[-it_0 \quad -it_0 \quad it_0 \quad it_0], \quad (12)$$

$$W^0=\begin{pmatrix} 0 & -it_0 & 0 & it_0 \\ -\frac{i}{t_0} & 0 & \frac{i}{t_0} & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \quad (13)$$

with $t_0=\sqrt{t^2+\varepsilon-1}$. Similarly, the transverse elements of the electromagnetic field mode have the same form as Eq. (11), except for $W$, $\gamma$, $A_j$ being replaced by $W^0$, $\gamma^0$, $C_j$.

For the requirement of the surface modes, $A_3=A_4=B_1=B_2=0$. Imposing the boundary conditions that the transverse elements of the electromagnetic field are continuous at $z=0$ and $a$. Namely, we will get a system of eight linear homogeneous equations relating the unknown coefficients $A_1$, $A_2$, $B_3$, $B_4$, $C_1$, $C_2$, $C_3$, $C_4$. It has the nontrivial solutions under the condition that the determinant of its coefficients is equal to zero, which lead to the equation for the determination of the proper frequency $\omega$

$$\begin{vmatrix} \frac{t+t_0}{t} & \frac{t-t_0}{t} & 1+\frac{t_0}{t_1} & 1-\frac{t_0}{t_2} \\ \frac{t-t_0}{t} & \frac{t+t_0}{t}y & 1-\frac{t_0}{t_1} & \left(1+\frac{t_0}{t_2}\right)y \\ \frac{\left(\varepsilon+\frac{t}{t_0}\right)\tan\theta}{t^2} & \frac{\left(\varepsilon-\frac{t}{t_0}\right)\tan(\varphi+\theta)}{t^2} & \cot\theta\left(1+\frac{t^2}{\varepsilon t_0 t_1}\right) & \cot(\varphi+\theta)\left(1-\frac{t^2}{\varepsilon t_0 t_2}\right) \\ \frac{\left(\varepsilon-\frac{t}{t_0}\right)\tan\theta}{t^2} & \frac{\left(\varepsilon+\frac{t}{t_0}\right)\tan(\varphi+\theta)}{t^2}y & \cot\theta\left(1-\frac{t^2}{\varepsilon t_0 t_1}\right) & \cot(\varphi+\theta)\left(1+\frac{t^2}{\varepsilon t_0 t_2}\right)y \end{vmatrix}=0 \quad (14)$$

with $y=e^{2K_0 t_0 a}$ and $t_2=\sqrt{t^2+t^2\delta\cos^2(\theta+\varphi)-\varepsilon\delta\sin^2(\theta+\varphi)}$.

In order to compare with the Lifshitz result for isotropic dielectrics, we assume that $\delta$ is a small variable, $\delta\ll 1$. From Eq. (14), we obtain, after some simple but lengthy algebra,

$$(1-Q^{\perp 2})_{\omega_N^{\perp}}(1-Q^{\parallel 2})_{\omega_N^{\parallel}}=0. \quad (15)$$

The superscripts $\parallel$ and $\perp$ indicate the two polarizations and

$$Q^{\perp 2}=\frac{(K_3-K_2)^2}{(K_3+K_2)^2}[1+a_1\delta+a_2\delta^2+o(\delta^3)]e^{-2K_3 a}, \quad (16)$$

$$Q^{\parallel 2}=\frac{(K_3\varepsilon-K_2)^2}{(K_3\varepsilon+K_2)^2}[1+b_1\delta+b_2\delta^2+o(\delta^3)]e^{-2K_3 a}, \quad (17)$$

with

$$a_1 = \frac{K_3\varepsilon[2-\cos 2\theta-\cos 2(\varphi+\theta)]}{2K_2(\varepsilon-1)}, \tag{18}$$

$$b_1 = \frac{K_3K_2\varepsilon[2+\cos 2\theta+\cos 2(\varphi+\theta)]}{2(K_3^2\varepsilon^2-K_2^2)}, \tag{19}$$

$$a_2 = \frac{K_3^2\varepsilon^2\cos 2\varphi}{8K_2^2(\varepsilon-1)^2} + \frac{K_3}{16K_2}\frac{(K_3^2-K_2^2)\varepsilon^2\cos 2\varphi}{(K_3^2\varepsilon-K_2^2)(\varepsilon-1)^2} + C_1 + f_1(\varphi,\theta), \tag{20}$$

$$b_2 = \frac{K_3^2K_2^2\varepsilon^2\cos 2\varphi}{8(K_3^2\varepsilon^2-K_2^2)^2} - \frac{K_3}{16K_2}\frac{(K_3^2-K_2^2)\varepsilon^2\cos 2\varphi}{(K_3^2\varepsilon-K_2^2)(\varepsilon-1)^2} + C_2 + f_2(\varphi,\theta), \tag{21}$$

$$K_3^2 = \frac{\omega^2}{c^2}(t^2+\varepsilon-1), \quad K_2^2 = \frac{\omega^2}{c^2}t^2. \tag{22}$$

The expressions of $C_1$, $C_2$, $f_1(\varphi,\theta)$, $f_2(\varphi,\theta)$ are very complicated, and $f_1(\varphi,\theta)$, $f_2(\varphi,\theta)$ are period functions of $\theta$. Because they do not contribute to the last moment $\tau(\varphi)$ depending on $\varphi$, here we do not show them.

Introducing $G^{\perp}(\omega)=1-Q^{\perp^2}$ and $G^{\parallel}(\omega)=1-Q^{\parallel^2}$, we rewrite Eq. (15) as

$$G^{\perp}(\omega_N^{\perp})G^{\parallel}(\omega_N^{\parallel}) = 0. \tag{23}$$

The electromagnetic energy associated with the mode $(\vec{k},\omega_N^{\perp})$ and $(\vec{k},\omega_N^{\parallel})$ are the zero-point energy $\hbar\omega_N^{\perp}/2$ and $\hbar\omega_N^{\parallel}/2$, that is,

$$\langle 0|\frac{1}{8\pi}\int d\vec{r}\left(\frac{d(\varepsilon\omega)}{d\omega}|\vec{E}_q|^2+|\vec{B}_q|^2\right)_{\omega_N^{\perp}}|0\rangle = \frac{1}{2}\hbar\omega_N^{\perp}, \tag{24}$$

$$\langle 0|\frac{1}{8\pi}\int d\vec{r}\left(\frac{d(\varepsilon\omega)}{d\omega}|\vec{E}_q|^2+|\vec{B}_q|^2\right)_{\omega_N^{\parallel}}|0\rangle = \frac{1}{2}\hbar\omega_N^{\parallel}, \tag{25}$$

where $|0\rangle$ represents the vacuum photon state.

## III. CASIMIR TORQUE IN VACUUM

We first calculate the total zero-point energy per unit area $E$ associated with the quantized surface modes discussed in the preceding section. The total energy of the system is simply the sum of the zero-point energies of the surface modes,

$$V = \sum_{\vec{k},N}\frac{1}{2}\hbar\omega_N^{\perp} + \sum_{k,N}\frac{1}{2}\hbar\omega_N^{\parallel}. \tag{26}$$

Sum over $\vec{k}$ in the above expression is performed as in Ref. [24] and the zero-point energy per unit area can be expressed as

$$E = \frac{\hbar}{8\pi^2}\int_0^{\infty}dkk\int_0^{2\pi}d\theta\left(\sum_N\omega_N^{\perp}+\sum_N\omega_N^{\parallel}\right). \tag{27}$$

Sums over $N$ can be performed with the help of the argument theorem as applied in Refs. [24,25], which results in

$$E(a,\varphi) = \frac{\hbar}{8\pi^3}\int_0^{\infty}dkk\int_0^{2\pi}d\theta\int_0^{\infty}d\xi[\ln G^{\perp}(i\xi)+\ln G^{\parallel}(i\xi)] \tag{28}$$

with the variable change $\omega=i\xi$.

Equation (28) may be rearranged as some known results. Following Lifshitz [14], we introduce a variable $p$, defined by

$$k^2 = \varepsilon_3\frac{\xi^2}{c^2}(p^2-1). \tag{29}$$

We will obtain

$$E(a,\varphi) = \frac{\hbar}{8\pi^3c^2}\int_1^{\infty}pdp\int_0^{2\pi}d\theta\int_0^{\infty}\xi^2d\xi[\ln G^{\perp}(i\xi) + \ln G^{\parallel}(i\xi)]. \tag{30}$$

At large distances $a\gg\lambda_0$ (where $\lambda_0$ is the characteristic absorption wavelength of the semispace dielectric matter), the dielectric permittivities can be represented by their static values at $\xi=0$, $\varepsilon_{10}=\varepsilon_{20}=\varepsilon(0)$. Introducing in Eq. (30) the variable $x=2p\xi a/c$, one obtains

$$E(a,\varphi) = -\frac{\hbar c\pi}{30a^3}\frac{5}{32\pi^4}\int_1^{\infty}\frac{dp}{p^2}\int_0^{2\pi}d\theta \times\int_0^{\infty}x^3dx\left(\frac{1}{\frac{1}{Q^{\perp^2}}-1}+\frac{1}{\frac{1}{Q^{\parallel^2}}-1}\right). \tag{31}$$

If $\delta=0$, Eq. (31) returns back to the well-known Lifshitz result in the case of two same semi-infinite isotropic dielectric plates [1,14]. For small $\delta$, Eq. (31) can be expanded as

$$E(a,\varphi) \approx E_0(a) - \frac{\pi^2\hbar c}{720a^3}\Psi(\varepsilon_0)\delta^2\cos 2\varphi, \tag{32}$$

where $E_0(a)$ represents the sum of elements not depending on $\varphi$ in $E(a,\varphi)$, and the second term is associated with the angle $\varphi$, which introduces a vacuum torque per unit area,

$$\tau(\alpha) = -\frac{\partial E(a,\varphi)}{\partial\varphi} \approx -\frac{\pi^2\hbar c}{360a^3}\Psi(\varepsilon_0)\delta^2\sin 2\varphi. \tag{33}$$

The relation between $\Psi$ and $\varepsilon$ can be expressed as

$$\Psi(\varepsilon) = \frac{15}{2\pi^4}\int_1^{\infty}\frac{dp}{p^2}\int_0^{\infty}x^3dx\left(\frac{y_1(1+y_1)}{(1-y_1)^3}\frac{p^2\varepsilon^2}{8(p^2+\varepsilon-1)(\varepsilon-1)^2} + \frac{y_2(1+y_2)}{(1-y_2)^3}\frac{p^2(p^2+\varepsilon-1)\varepsilon^2}{8[p^2(\varepsilon+1)-1]^2(\varepsilon-1)^2} + \frac{\varepsilon^2p^2e^{-x}}{4[\sqrt{(p^2+\varepsilon-1)}+p]^2[\sqrt{(p^2+\varepsilon-1)}+p\varepsilon]^2} \times\frac{1-y_1-y_2}{(1-y_1)^2(1-y_2)^2}\right) \tag{34}$$

with

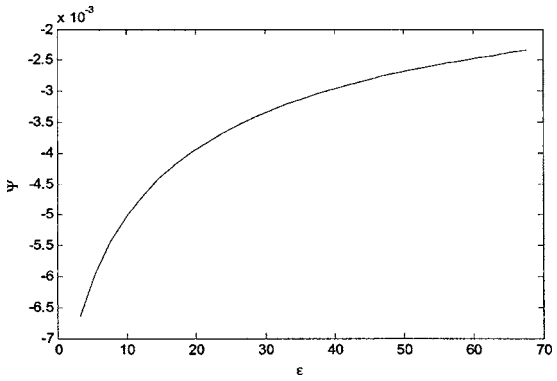



FIG. 3. Numerical calculation of function $\Psi(\varepsilon)$.

$$y_1 = \left(\frac{\sqrt{p^2+\varepsilon-1}-p}{\sqrt{p^2+\varepsilon-1}+p}\right)^2 e^{-x}$$

$$\text{and } y_2 = \left(\frac{\sqrt{p^2+\varepsilon-1}-p\varepsilon}{\sqrt{p^2+\varepsilon-1}+p\varepsilon}\right)^2 e^{-x},$$

which can be plotted as shown in Fig. 3.

From Eq. (33), we know that the torque varies as $\sin(2\varphi)$ and has maximum magnitude at $\varphi=\pi/4$ and $\varphi=3\pi/4$, which is coincident with Refs. [10,23]. As an example, we consider two 10 cm diameter disks made out of ZnO. When they are placed in vacuum with a distance of 1 $\mu$m apart, the maximum torque will be about $1\times10^{-15}$ N m, since ZnO has $\varepsilon_{\|}=9.9, \varepsilon_{\perp}=8.4$, corresponding $\delta=0.18$. This torque could be measured in the future torsion balance experiment in our laboratory.

## IV. CONCLUSIONS

We have performed detailed calculations of the vacuum energy between two birefringent plates using the quantized surface mode technique. And an analytical approximate expression of Casimir torque was obtained in the retarded limit, which is found to vary as $\sin(2\varphi)$ with $\varphi$, the angle between the two optical axes. This result is a supplement to the numerical calculation obtained by Munday *et al.* [23].

## ACKNOWLEDGMENTS

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